

**THE EXACT BOUND-STATE ANSAETZE
AS ZERO-ORDER APPROXIMATIONS IN PERTURBATION THEORY
I. THE FORMALISM AND PADÉ OSCILLATORS**

M. Znojil

Institute of Nuclear Physics, Czechosl. Acad. Sci., 250 68 Řež near Prague, Czechoslovakia

A new perturbative prescription is proposed. It differs from the textbook Rayleigh-Schroedinger (RS) theory by the assumption of the user's incomplete knowledge of the unperturbed spectra. We only need a single eigenvalue and eigenstate of H_0 . In our quasi-RS (QRS) formulas, this eigenstate keeps its "unperturbed" RS interpretation, while the role of the (non-available) RS unperturbed spectrum is taken over by certain auxiliary matrix continued fractions. These quantities represent a weaker though entirely sufficient "compressed" information about H_0 and, after a suitable partitioning, enable one to combine the usual (i.e., sufficiently simple, here: harmonic-oscillator) working basis $|n\rangle$ with a very broad (here: arbitrary Padé-anharmonic) class of the zero-order Hamiltonians H_0 .

1. Introduction

Exact solutions of Schroedinger equations often mediate our qualitative understanding of a quantum system. Unfortunately, only very few Hamiltonians admit an exact solution. Hence, even a finite set of elementary states, say of the type

$$\psi_{0(j)}(r) = r^{l+1} \times \text{polynomial}(r) \times \exp[\text{polynomial}(r)], \quad j = 1, 2, \dots, s \quad (1.1)$$

may be of interest, especially whenever it still represents an (incomplete) exact solution to some "unperturbed" bound-state problem

$$H_0|\psi_{0(j)}\rangle = E_{0(j)}|\psi_{0(j)}\rangle. \quad (1.2)$$

With the standard specification of parity ($l = -1$ and 0) or angular momentum ($l = 0, 1, \dots$) in the respective one- or three-dimensional Hamiltonians

$$H_0 = -\frac{d^2}{dr^2} + \frac{l(l+1)}{r^2} + V_0(r) \quad (1.3)$$

the partial exact solvability of the quantum non-harmonic oscillators has been discovered by Singh et al. [1] and by Flessas et al. [2] during their studies of the

potentials

$$V_0(r) = ar^2 + br^4 + r^6, \quad b = b(a) \quad (1.4)$$

and

$$V_0(r) = r^2 + \lambda r^2/(1 + gr^2), \quad \lambda = \lambda(g). \quad (1.5)$$

An extension of these constructions to arbitrary polynomial [3] and Padé [4] forces and also to the Schroedinger-like equations in curved spaces [5–7] and to the many-body systems [5, 6] proved quite straightforward.

Here, we shall restrict our attention to the rational or Padé anharmonicities which generalize (1.5),

$$V_0(r) = r^2 + \sum_{m=0}^p f_m r^{2m} / G(r),$$

$$G(r) = \sum_{n=0}^p g_n r^{2n} > 0, \quad g_0 = 1, \quad g_p > 0. \quad (1.6)$$

We shall complement the various existing (e.g., computational, phenomenological and methodical) consequences of existence of the generalized oscillator states (1.1) [7, 8] by a proposal and description of their use within the framework of perturbation theory. Indeed, no finite number of these elementary particular solutions may help as an ordinary unperturbed basis: In the standard Rayleigh-Schroedinger (RS) perturbative prescriptions, $s = \infty$ would be needed [9]. At the same time, the generalized RS (GRS) formalism of our preceding paper [10] does not require any explicit zero-order solutions at all ($s = 0$ is assumed). Hence, an appropriate intermediate or quasi-RS (QRS) form of perturbation prescription is called for.

The first two sections of our present paper will deliver some preliminaries. For the sake of definiteness, the arbitrary Padé-anharmonic oscillator (1.5) is considered – a comparison of our results with the alternative GRS perturbative treatment of the same example [10] is being kept in mind. In this context, Section 2 will review some basic properties (viz., $2p + 1$ – diagonal structure and possible partitioning and truncation) of our matrix Schroedinger equation. In Section 3, the importance and feasibility of constructions of the suitable Padé – anharmonic zero – order Hamiltonians H_0 at any $p \geq 1$ is emphasized and given a detailed explanation.

In contrast to the universal GRS scheme, our present QRS prescription takes all the available information about the Padé-anharmonic H_0 's into account. Section 4 presents the core of the idea: The RS-type hierarchy of equations is shown to define all the QRS corrections. In Section 5, each of these definitions is then shown to be reducible to the solution of a finite system of linear algebraic equations.

An auxiliary partitioning and an introduction of related matrix continued fractions. (Appendix A) proves essential for a simplification of the explicit QRS formulas. Section 6 is devoted to the pertaining discussion – the energy corrections are given by a closed formula, and the re-normalization freedom is shown to be restricted to a few components of the wave-functions.

The more detailed analysis of the separate QRS corrections is moved to our forthcoming paper II. Also the explicit numerical tests are postponed to this second half of the series. Here, only a short summary of the general QRS scheme is added in Section 7.

2. Padé oscillators and their non-symmetric matrix Schroedinger equations

The coupling f_p in Eq. (1.6) fixes just the (irrelevant) zero of the energies. Hence, a change of notation

$$\begin{aligned} V(r) &= r^2 + f_p/g_p + Z(r)/G(r), \\ Z(r) &= \sum_{n=0}^{p-1} z_n r^{2n}, \quad z_n = f_n - f_p g_n/g_p \end{aligned} \quad (2.1)$$

leads to the Padé-oscillator Schroedinger equation in the form

$$\begin{aligned} [(H_{\text{HO}} - F) G(r) + Z(r)] \varphi(r) &= 0, \quad F = E - f_p/g_p, \\ \varphi(r) &= G^{-1}(r) \psi(r). \end{aligned} \quad (2.2)$$

Here, H_{HO} denotes the ordinary harmonic oscillator Hamiltonian H_0 with the $p = 0$ potential $V_0 = V_{\text{HO}} = r^2$.

In terms of the complete set of eigenstates of H_{HO} ,

$$H_{\text{HO}}|n\rangle = e_n|n\rangle, \quad e_n = 4n + 2l + 3, \quad n = 0, 1, \dots \quad (2.3)$$

we may expand

$$\varphi(r) = \sum_{n=0}^N \langle r | n \rangle \langle n | \varphi \rangle, \quad N \rightarrow \infty \quad (2.4)$$

and employ the well-known formulas for the matrix elements [11],

$$\begin{aligned} \langle n | r^2 | n \rangle &= 2n + l + 3/2, \\ \langle n | r^2 | n+1 \rangle &= \langle n+1 | r^2 | n \rangle = [(n+1)(n+l+3/2)]^{1/2}, \quad n = 0, 1, \dots \end{aligned} \quad (2.5)$$

As a consequence, our Schroedinger equation acquires the infinite-dimensional matrix form

$$\begin{aligned} T|\varphi\rangle &= EG|\varphi\rangle, \\ T &= (H_{\text{HO}} + f_p/g_p) G(r) + Z(r) \end{aligned} \quad (2.6)$$

of a generalized eigenvalue problem [12].

We may also introduce a truncated and partitioned form of Eq. (2.6), with

$$R = T - EG = \begin{pmatrix} A_1 & B_1 & & & \\ C_2 & A_2 & B_2 & & \\ & \dots & \dots & \dots & \\ & & C_{t-1} & A_{t-1} & B_{t-1} \\ & & & C_t & A_t \end{pmatrix}, \quad (2.7)$$

where $\dim A_2 = \dim A_3 = \dots = \dim A_{t-1} = p$. The two partitions remain variable, $\dim A_1 = p_1$, $\dim A_t = p_t$. Such a freedom may prove useful, e.g., for an unrestricted variability of the total dimension of R , $\dim R = p_1 + (t-2)p + p_t$.

It is necessary to emphasize that the $2p+1$ -diagonal matrix T is not symmetric – this means that the admissibility of the truncation of Eq. (2.6) is not obvious. Fortunately, the correct behaviour of the $t \rightarrow \infty$ limit may be confirmed by the “fixed-point” method of Ref. [13]. This leaves us free to treat Eq. (2.6) as an infinite-dimensional limit of its finite subsets (cf. also Ref. [10]).

3. The particular exact and elementary bound states of the general Padé-anharmonic oscillator

Our Ansatz (1.1) is equivalent to termination of a general infinite series, say, (2.4),

$$|\varphi_0\rangle = \sum_{n=0}^N |n\rangle \langle n | \varphi_0\rangle, \quad N = N_0 < \infty. \quad (3.1)$$

Obviously, in the spirit of the $p=1$ idea of Whitehead et al. [14], this reduces Eq. (2.6) to mere $N+p+1$ linear algebraic equations

$$\sum_{n=0}^N \langle m | R_0 | n \rangle \langle n | \varphi_0 \rangle = 0, \quad m = 0, 1, \dots, N+p. \quad (3.2)$$

Of course, such an overcomplete system of equations (for $N+1$ unnormalized components of $|\varphi_0\rangle$) imposes also some $p+1$ independent restrictions upon matrix elements of $R = R_0$ itself.

The first and simplest restriction on R_0 is represented by the last, $(N+p)$ -th row in (3.2). It contains no contribution from the $2p-1$ -diagonal matrix $Z(r)$ and fixes the value

$$E_0 = f_p/g_p + e_{N+p} \quad (3.3)$$

of the “unperturbed” energy (cf. also Ref. [4]).

The next N rows of (3.2)

$$\langle m | R_0 | \varphi_0 \rangle = 0, \quad m = p+N-1, \quad p+N-2, \dots, p \quad (3.4)$$

have a recurrent interpretation: Each of them defines the next component of our wavefunction,

$$\begin{aligned} \langle N-m | \varphi_0 \rangle &= -(\langle p+N-m | R_0 | N-m \rangle)^{-1} \sum_{k=1}^K \langle p+N-m | R_0 \\ &\times | N-m+k \rangle \langle N-m+k | \varphi_0 \rangle \\ K &= \min(m, 2p), \quad m = 1, 2, \dots, N. \end{aligned} \quad (3.5)$$

This procedure starts from a nonzero normalization $\langle N | \varphi_0 \rangle \neq 0$.

The remaining p rows of (3.2) must still be compatible with the nontriviality of solution $|\varphi_0\rangle$. This leads to the p independent conditions

$$\det R_0\{j\} = 0, \\ (R_0\{j\})_{mn} = \langle m+j-1 | R_0 | n \rangle, \quad m, n = 0, 1, \dots, N, \quad j = 1, 2, \dots, p. \quad (3.6)$$

The main merit of such a choice and notation lies in the obvious existence of some p independent bra-vectors $\langle \chi\{j\} |$ such that

$$\langle \chi\{j\} | R_0\{j\} = 0 \quad (3.7)$$

and

$$\langle \chi\{j\} | R_0 = \langle \chi\{j\} | R_0 Q, \\ Q = \sum_{m=N+1}^{N+2p-1} |m\rangle \langle m|. \quad (3.8)$$

These auxiliary finite-dimensional vectors will be needed later.

4. Perturbation theory

Let us now assume that the partially solvable interaction $V_0(r)$ becomes modified by an arbitrary perturbation proportional to a small parameter h . The variable h may represent a simple change of a coupling in Eq. (2.1)¹⁾ as well as any more general measure of difference between V and V_0 [16]. As a consequence, the related $2p+1$ -diagonal matrices G_0 and T_0 become replaced by the expansions

$$G = G_0 + hG_1 + h^2G_2 + \dots \\ T = T_0 + hT_1 + h^2T_2 + \dots \quad (4.1)$$

with some h -independent matrix coefficients. The available particular solutions of Eq. (1.2) or (3.2) may be then taken as the most natural zero-order approximants to the perturbed bound states. In Ref. [10], such a possibility has not been noticed at all.

In the first step of our QRS perturbative considerations, let us put

$$E = E(h) = E_0 + hE_1 + h^2E_2 + \dots \quad (4.2)$$

and combine this series with our assumption (4.1). As a result, the operator R in (2.7) also acquires an infinite-series form

$$R = R(h) = R_0 + hR_1 + h^2R_2 + \dots = \\ = (T_0 - E_0G_0) + h(T_1 - E_0G_1 - E_1G_0) + \dots \quad (4.3)$$

Moreover, the parallel Ansatz for the wavefunctions

$$|\varphi\rangle = |\varphi_0\rangle + h|\varphi_1\rangle + h^2|\varphi_2\rangle + \dots \quad (4.4)$$

¹⁾ See Section 4 of the Part II of this paper, [15].

converts the full, "perturbed" Schrodinger equation (2.6) into the infinite RS-type hierarchy of relations

$$\begin{aligned}
 (T_0 - E_0 G_0) |\varphi_0\rangle &= 0, \\
 (T_0 - E_0 G_0) |\varphi_1\rangle + (T_1 - E_0 G_1 - E_1 G_0) |\varphi_0\rangle &= 0 \\
 (T_0 - E_0 G_0) |\varphi_k\rangle + (T_1 - E_0 G_1 - E_1 G_0) |\varphi_{k-1}\rangle + \dots \\
 \dots + (T_k - E_0 G_k - \dots - E_{k-1} G_1 - E_k G_0) |\varphi_0\rangle &= 0
 \end{aligned} \tag{4.5}$$

which are ordered in accord with ascending powers of our small parameter $h \neq 0$.

Each of the QRS equations (4.5) may now be understood as an implicit definition of the k -th corrections in (4.2) and (4.4). At any order k , the k -th row of Eq. (4.5) acquires the same, universal form

$$R_0 |\varphi_k\rangle - E_k G_0 |\varphi_0\rangle = |\tau\rangle. \tag{4.6}$$

Here, the k -dependent abbreviation $|\tau\rangle$ contains just the "known" or "old" quantities $|\varphi_{k-1}\rangle, \dots, |\varphi_0\rangle$ and $E_{k-1}, \dots, E_0$.

5. The finite-dimensional form or recurrences (4.6)

A. The energy-independent components of the wavefunctions

Let us choose, say, $p = 3$ and $N = 5$ and visualize R_0 as an infinite-dimensional matrix with the banded structure,

$$\begin{array}{cccc|cccc|cccc|}
 y & x & x & x & & & & & & & & & \\
 x & y & x & x & x & & & & & & & & \\
 x & x & y & x & x & x & & & & & & & \\
 x & x & x & y & x & x & x & & & & & & \\
 & x & x & x & y & x & x & x & & & & & \\
 & & x & x & x & y & x & x & x & & & & \\
 \hline
 & & & x & x & x & y & x & x & x & & & \\
 & & & & x & x & x & y & x & x & x & & \\
 & & & & & x & x & y & x & x & x & & \\
 & & & & & & x & x & x & y & x & x & x \\
 & & & & & & & x & x & x & y & x & x & x \\
 \hline
 & & & & & & & x & x & x & y & x & x & x \\
 & & & & & & & & x & x & x & y & x & x & x \\
 & & & & & & & & & \dots & & & &
 \end{array} \tag{5.1}$$

Here, the nonzero elements are denoted by x and, on the main diagonal, y . The pairs of auxiliary lines indicate the range of the projector Q (see (3.8)). The solid

line also specifies the boundary of the range of the second projector

$$S = \sum_{m=N+2p}^{\infty} |m\rangle \langle m|. \quad (5.2)$$

The band-matrix structure of the denominator G_0 implies that $SG_0 P|\varphi_0\rangle = 0$. Hence, the S -projection of our basic relations (4.6) does not contain the energy correction E_k ,

$$S R_0 |w\rangle + S R_0 |v\rangle = S |\tau\rangle \quad (5.3)$$

$$|v\rangle = Q|\varphi_k\rangle, \quad |w\rangle = S|\varphi_k\rangle$$

and has a matrix-continued-fractional (MCF) solution

$$|w\rangle = (SR_0 S)^{-1} (S|\tau\rangle - S R_0 |v\rangle) \quad (5.4)$$

which is described in more detail in Appendix A below.

The MCF formula (5.4) defines the infinite-dimensional part $|w\rangle$ of $|\varphi_k\rangle$ as a vector proportional to the "model-space" or Q -projection $|v\rangle$ of $|\varphi_k\rangle$. We may insert such a formula in all the rows of (4.6) where it is coupled to the remaining components of the wavefunction corrections,

$$\langle m| R_0 |v\rangle + \langle m| R_0 |w\rangle = \langle m| \tau \rangle \quad (5.5)$$

$$m = N + 2p - 1, N + 2p - 2, \dots, N + p + 1.$$

Now, the price we pay for the elimination of the infinite-dimensional $|w\rangle$'s may be seen in the presence of the MCF coefficients in our definitions of the QRS corrections.

B. The refinement of partitioning and MCF formulas

In the way inspired again by the illustration (5.1), let us now replace the partitioning (2.7) of R_0 by its refinement

$$R_0 = \left[\begin{array}{c|c|c} A_{(0)} & B_{(0)} & \\ \hline C_{(1)} & A_{(1)} & B_{(1)} \\ \hline & C_2 & A_2 \ B_2 \\ & & C_3 \ A_3 \ \dots \\ & & \dots \end{array} \right], \quad (5.6)$$

where $\dim A_{(0)} = N + 1$ and $\dim A_{(1)} = 2p - 1$, while $\dim A_2 = \dim A_3 = \dots = p$ as before. Notice that $A_{(1)} = QR_0Q$ and that the possible confusion is being avoided by our writing some of the partition-subscripts in parentheses.

In the light of Appendix A, Eq. (5.4) has an explicit MCF form as follows,

$$\begin{aligned}
 |w\rangle = \begin{pmatrix} |w_2\rangle \\ |w_3\rangle \\ |w_4\rangle \end{pmatrix} = - \begin{pmatrix} 1 \\ -F_3 & C_3 \\ F_4 & C_4 & F_3 & C_3 \end{pmatrix} F_2 & C_2 |v\rangle \\
 + (XS)^{-1} \times \begin{pmatrix} F_2 \\ F_3 \\ \dots \end{pmatrix} \times (SY)^{-1} \times \begin{pmatrix} |\tau_2\rangle \\ |\tau_3\rangle \\ \dots \end{pmatrix}.
 \end{aligned} \quad (5.7)$$

Its insertion also leads to the final and compact form

$$\begin{aligned}
 \langle m | (A_{(1)} - B_{(1)} F_2 C_2) | v \rangle &= \langle m | (|\tau_{(1)}\rangle - B_{(1)} F_2 | \varrho \rangle), \\
 | \varrho \rangle &= |\tau_2\rangle - B_2 F_3 [|\tau_3\rangle - B_3 F_4 (|\tau_4\rangle - \dots)] \\
 m &= N + 2p - 1, \quad N + 2p - 2, \dots, N + p + 1
 \end{aligned} \quad (5.8)$$

of our remaining E_k -independent subset (5.5) of equations (4.6).

We may summarize: The $p - 1$ equations (5.8) plus the first $N + p + 1$ $\langle m |$ -projections of (4.6) form our final and finite set of linear algebraic equations for the k -th QRS corrections E_k and $|\varphi_k\rangle$. In principle, they may be solved numerically.

6. Discussion

A. An elimination of the energies

As long as the energy variable E_k enters just the first $m = N + p + 1$ items ($\langle m |$ -projections) of Eq. (4.6), we may take the last row of this set, i.e., the $(N + p)$ -th projection

$$\langle N + p | R_0 | v \rangle - \langle N + p | G_0 | \varphi_0 \rangle E_k - \langle N + p | \tau \rangle = 0, \quad (6.1)$$

as the explicit definition of energies or as an abbreviation

$$E_k = \alpha + \langle \beta | v \rangle. \quad (6.2)$$

It is important to notice that all our equations (rows of (4.6)) remain linear in $|v\rangle$ after the insertion of (6.2).

B. A separation of the renormalization ambiguity

In the GRS perturbation algorithm of Ref. [10], even the zero-order approximants were constructed in terms of the auxiliary MCF quantities. In such a setting, the rule $\det R_0 = 0$ and the related re-normalization freedom

$$|\varphi_k\rangle \rightarrow |\varphi_k\rangle + \text{const} |\varphi_0\rangle \quad (6.3)$$

caused troubles. Indeed, the GRS transformation or ambiguity (6.3) modified all the components of wavefunctions. In the present QRS formalism, on the contrary,

only the first $N + 1$ components of $|\varphi_k\rangle$ (i.e., the vector $|w_{(0)}\rangle$ in the partitioned notation of Eq. (5.6)) are involved in (6.3).

In each ($=k$ -th) step of the construction of corrections we may use the set of vectors $\langle\chi\{j\}|$, $j = 1, 2, \dots, p$ which were introduced in Section 3. In the light of Eq. (3.8), the action of these vectors on our basic Eq. (4.6) cannot contain the $\langle m|$ -projections of $|\varphi_k\rangle$ with $m < N + 1$ (i.e., the components which depend on the re-normalization (6.3)). Thus, we have the set of p new equations

$$\langle\chi\{j\}| R_0|v\rangle = \langle\chi\{j\}| \tau\rangle + \langle\chi\{j\}| G_0|\varphi_0\rangle (\alpha + \langle\beta| v\rangle), \quad j = 1, 2, \dots, p. \quad (6.4)$$

Obviously, in combination with the $p - 1$ equations (5.8), they form our final reduced QRS set of the $2p - 1$ linear algebraic equations for the $2p - 1$ unknown components of the vector $|v\rangle$.

C. The remaining components of the wavefunctions

In terms of the above, arbitrarily normalized vectors (solutions $|v\rangle = |w_{(1)}\rangle$ of the joint and linear algebraic set of Eqs. (5.8) + (6.4)), we may now define all the higher and renormalization- and energy-independent subvectors $|w_2\rangle, |w_3\rangle, \dots$ of $|\varphi_k\rangle$ by means of Eq. (5.7). At the same time, the last N -plet of independent equations (namely, an arbitrary subset of $\langle m|$ -projections of (4.6) with $m \leq N + p + 1$) still remains at our disposal. Of course, it has to define the last, renormalization- and energy-dependent subvector $|w_{(0)}\rangle$.

In a way similar to our specification (3.4) of $|\varphi_0\rangle$'s, we may again recommend the choice of the rows which form the transparent set of recurrences,

$$\begin{aligned} \sum_{i=m-p}^N \langle m| R_0|i\rangle \langle i| \varphi_k\rangle &= - \langle m| R_0|v\rangle \\ &+ \langle m| \tau\rangle + \langle m| G_0|\varphi_0\rangle (\alpha + \langle\beta| v\rangle) \\ m &= p + N - 1, p + N - 2, \dots, p. \end{aligned} \quad (6.5)$$

On the left-hand side, it contains the matrix $R_0\{j\}$ with $j = p$ as defined in (3.6), and its solution starts again from an arbitrary choice of the re-normalization $\langle N - \varphi_k\rangle \neq 0$. Step by step, we have

$$\begin{aligned} \langle N - m| \varphi_k\rangle &= \\ &= (\langle p + N - m| R_0|N - m\rangle)^{-1} \left(- \sum_{k=N-m+1}^{N-m+2p} \langle p + N - m| R_0|k\rangle \langle k| \varphi_k\rangle \right. \\ &\quad \left. + \langle N - m| \tau\rangle + \langle N - m| G_0|\varphi_0\rangle (\alpha + \langle\beta| v\rangle) \right), \quad m = 1, 2, \dots, N. \end{aligned} \quad (6.6)$$

The components $|v\rangle$ are required as an input again, and our construction of all the explicit QRS corrections is completed.

7. Summary

Of course, the numerical efficiency of the present QRS perturbation scheme will depend on the closeness of the Hamiltonian H to its approximation H_0 . A few illustrations and a more thorough analysis of the partially solvable examples will be presented in the forthcoming part II of the present series.

Here, we may only summarize the general formalism as a procedure consisting of the linear algebraic equations for $|v\rangle$'s (cf. Eqs. (5.8) and (6.4)), accompanied by the definitions of $|w\rangle$ (= the part of the wavefunction which is independent of the re-normalization, cf. Eq. (5.7)), of the energies (cf. (6.1) or (6.2)) and of the remaining part of the wavefunction (Eq. (6.6)) which depends on the re-normalization in an explicit manner.

Appendix A. An introduction of the matrix continued fractions

After partitioning and truncation (2.7), our Schroedinger equation (2.6) may be solved not only by the GRS perturbation method of Ref. [10], but also by an arbitrary specialized numerical technique [12]. Always, the related expansion

$$\psi(r) = \sum_{\substack{m,n=0 \\ |m-n| \leq p}}^{M-1} \langle r | m \rangle \langle m | G | n \rangle \langle n | \varphi \rangle \quad (\text{A.1})$$

$$M = p_1 + (t-2)p + p_t \rightarrow \infty$$

mediates a return to the original wavefunctions.

An improved understanding of the above, purely numerical procedures may be based on the factorization of R ,

$$R = Y \times \begin{pmatrix} 1/F_1 & & & \\ & 1/F_2 & & \\ & & \dots & \\ & & & 1/F_t \end{pmatrix} \times X. \quad (\text{A.2})$$

In accord with Ref. [10], we may employ here the pair of the triangular and block-two-diagonal factors

$$Y = \begin{pmatrix} 1 & B_1 F_2 & 0 & \dots & 0 \\ & 1 & B_2 F_3 & 0 & \dots \\ & & & \dots & \\ & & 1 & & B_{t-1} F_t \\ & & & & 1 \end{pmatrix} \quad (\text{A.3})$$

and

$$X = \begin{pmatrix} 1 & & & & \\ F_2 C_2 & 1 & & & \\ & & \dots & & \\ \dots & 0 & F_{t-1} C_{t-1} & 1 & \\ 0 & \dots & 0 & F_t C_t & 1 \end{pmatrix}. \quad (\text{A.4})$$

Indeed, the underlying matrix-continued-fractional (MCF) recurrences

$$F_i = 1/(A_i - B_i F_{i+1} C_{i+1}), \quad i = t, \quad t-1, \dots, 1, \quad t \rightarrow \infty \quad (\text{A.5})$$

with $F_{t+1} = 0$ convert (A.2) into an algebraic identity.

Let us now return to the illustration (5.1) (replacing R in (2.7) by R_0) and fix A_1 as the submatrix cut off by the pair of the solid lines, i.e., with $p_1 = N + 2p$. Then, the S -projected MCF factorization (A.2) may still be used and inverted,

$$(SR_0S)^{-1} = (XS)^{-1} \times \begin{pmatrix} F_2 & & \\ & F_3 & \\ & & \dots \\ & & & F_t \end{pmatrix} \times (SY)^{-1} \quad (\text{A.6})$$

since the formulas

$$(SY)^{-1} = \begin{pmatrix} 1 & -B_2F_3 & B_2F_3B_3F_4 & -B_2F_3B_3F_4B_4F_5 & \dots \\ & 1 & -B_3F_4 & B_3F_4B_4F_5 & \dots \\ & & & \dots & \\ & & & & 1 \end{pmatrix}$$

and, mutatis mutandis,

$$(XS)^{-1} = \begin{pmatrix} 1 & & \\ -F_3C_3 & 1 & \\ & & \dots \end{pmatrix}$$

provide the necessary new factors. Now, it remains for us to insert these formulas in the QRS definition (5.4) of $|w\rangle$. In contrast to the use of factorizations (A.2) in the trial and error search for energies in the numerical diagonalization algorithms [12], the QRS evaluation of auxiliary MCF quantities F_k must only be performed once, at a fixed value E_0 of our zero-order energy.

References

- [1] Singh V., Biswas S. N., Datta K.: Phys. Rev. D 18 (1978) 1901.
- [2] Flessas G. P.: Phys. Lett. A 72 (1979) 289; A 83 (1981) 121.
Flessas G. P., Das K. P.: Phys. Lett. A 78 (1980) 19.
- [3] Magyari E.: Phys. Lett. A 81 (1981) 116.
- [4] Znojil M.: J. Phys. A, Math. Gen. 16 (1983) 279.
- [5] Morozov A. J., Perelomov A. M., Rosly A. A., Shifman M. A., Turbiner A. V.: Int. J. Mod. Phys. A 5 (1990) 803.
- [6] Ushveridze A. G.: Lebedev Physics Institute Reports 1988, to appear in Soviet Physics (Allerton Press).
- [7] Chaudhuri R. N.: Phys. Rev. D 26 (1985) 3750.
Turbiner A. V.: Commun. Math. Phys. 118 (1988) 467.
Shifman M. A.: New findings in quantum mechanics, CERN-TH.5265/88 (preprint); Int. J. Mod. Phys. A 4 (1989) 2897.

- [8] Waken H.: *in* Laser Theory, Encyclopedia of Physics XXV/2. Van Nostrand, Princeton, 1970.
- Mitra A. K.: J. Math. Phys. 19 (1978) 2081.
- Itzykson C., Zuber J. B.: Quantum Field Theory. McGraw-Hill, New York, 1982.
- [9] Kumar K.: Perturbation Theory and the Nuclear Many-Body Problem. North Holland, Amsterdam, 1962.
- [10] Znojil M.: J. Math. Phys. 29 (1989) 2611; Czech. J. Phys. B, to appear.
- [11] Shiff L. I.: Quantum Mechanics. McGraw-Hill, New York, 1968.
- [12] Wilkinson J. H.: The Algebraic Eigenvalue Problem, Clarendon, Oxford, 1965.
- [13] Znojil M.: Czech. J. Phys. B 37 (1987) 1072.
- [14] Whitehead R. R., Watt A., Flessas G. P., Nagarajan M. A.: J. Phys. A, Math. Gen. 15 (1982) 1217.
- [15] Znojil M.: Czech. J. Phys. (to be published).
- [16] Bender C. M. et al.: Phys. Rev. B 37 (1988) 1472.